BINSHUAI WANG

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EDUCATION

George Washington University **Jan. 2022 – Present**
Ph.D. candidate in Computer Science
Supervisor: Peng Wei

University of California, Davis **Sep. 2019 – Dec. 2021**
M.S. in Computer Science

Beihang University **Sep. 2014 – Jul. 2018**
B.S. in Mathematics and Applied Mathematics
Hua Luogeng Honor Class

RESEARCH INTERESTS

Neural Network Optimization, Optimal Transport Theory, AI models (ViTs, LLMs), Hash Methods, Manifold Optimization, Numerical Optimization.

RESEARCH EXPERIENCE

George Washington University **Jan. 2022 – Present**
Washington, DC

- Proposed relative translation invariant Wasserstein distances (RW_p) to measure the discrepancy between probability distributions, with an efficient bi-level optimization framework; Developed RW_2-LP and RW_2-Sinkhorn algorithms to improve numerical stability for computing W_2 distance; Applied to a large-scale thunderstorm dataset to retrieve similar thunderstorm patterns.
- Proposed a probability-distribution framework to evaluate computer vision models by measuring discrepancies between the training dataset and operational design domain (ODD) samples; used diffusion models to densify ODD samples and mitigate data scarcity.
- Developed CNN/MLP-based Vision Transformers in PyTorch to encode sequential thunderstorm radar snapshots and identify similar thunderstorm events via latent-space retrieval.
- Proposed the Relative Wasserstein (RW_2) geometry, introducing notions of angles and orthogonal projections to the probability distribution space. Implemented distribution dynamic visualizations.

University of California, Davis **Nov. 2020 – Dec. 2021**
Davis, CA

- Developed an algorithm for the problem of generating high-dimensional cutting planes in integer programming. The algorithm recasts the problem into verification of the intersection status of a group of polyhedra pairs, and uses R tree with adaptive projection techniques to retrieve the intersected polyhedra pairs. The algorithm has been implemented in object-oriented programming (OOP) for Sagemath.

Beihang University **Sep. 2016 – Jul. 2019**
Beijing, China

- Proposed a new family of locality-sensitive hashing functions for the problem of subspace-to-subspace search applied in large-scale datasets. Those functions preserve the angular distances. Experiments show that the proposed approach outperforms the state-of-the-art methods, in terms of both accuracy and efficiency (up to 16× speedup); awarded Best Student Paper at PCM 2018.

PUBLICATIONS

- **Binshuai Wang**, Peng Wei. "Relative Wasserstein Angle and the Problem of the W_2-Nearest Gaussian Distribution," arXiv preprint arXiv:2601.22355, 2026.
- **Binshuai Wang**, Qiwei Di, Ming Yin, Mengdi Wang, Quanquan Gu, Peng Wei. "Relative-Translation Invariant Wasserstein Distance", Transactions on Machine Learning Research, 2026.
- **Binshuai Wang**, Can Vu, Peng Wei. "Performance Estimation of Aircraft Computer Vision Models using Optimal Transport-based Distributional Mismatch", **Best of Session Award**, IEEE Digital Avionics Systems Conference, 2025.
- **Binshuai Wang**, Peng Wei. "Thunderstorm Pattern Analysis Using Vision Transformers and Optimal Transport for Aviation Operations", AIAA AVIATION, 2025.
- Amber Liu, **Binshuai Wang**, Peng Wei. "Identifying Similar Weather Patterns Using Modified p-Wasserstein Distances", AIAA SciTech, 2024.
- **Binshuai Wang**, Peng Wei. "Identifying Similar Thunderstorm Sequences for Airline Decision Support via Optimal Transport Theory", AIAA SciTech, 2023.
- Yi Xu, Xianglong Liu, **Binshuai Wang**, Ke Xia, Xianbin Cao. "Fast Nearest Subspace Search via Random Angular Hashing", IEEE Transactions on Multimedia, vol. 23, pp. 342-352, 2021.
- **Binshuai Wang**, Xianglong Liu, Ke Xia, Kotagiri Ramamohanarao, Dacheng Tao. "Random Angular Projection for Fast Nearest Subspace Search", **Best Student paper**, 19th Pacific-Rim Conference on Multimedia (PCM), 2018.

SKILL COLLECTION

Languages: Python, C, C++, Git, CUDA, Pascal, html, Js, SQL, MongoDB, React.
Systems: Unix/Linux, Google Cloud, AWS, Azure.
Software/Packages: PyTorch/TensorFlow, PySpark, Pandas, Mathematica, Matlab, Gurobi/CPlex, Bluesky, Ray.

HONORS AND AWARDS

Best of Session Award, IEEE Digital Avionics Systems Conference (DASC)	2025
Graduate Research Award, George Washington University	2022-2026
Best Student Paper Award, Pacific-Rim Conference on Multimedia (PCM)	2018
Outstanding Undergraduate Thesis, Beihang University	2018
Honorable Mention, COMAP's Mathematical Contest in Modeling (MCM)	2017, 2016
Award for Excellence, Department of Mathematics, Beihang University	2016
Student Academic Excellence Award, Beihang University	2015